SE444 – Artificial Intelligence

Quiz 4: Bayesian Network Inference

Lecture 11: Bayesian Networks & Probabilistic Inference

Time: 30 minutes Total: 15 Questions

Name: _____ **ID:** _____

Instructions: Choose the best answer for each question. Some questions may have multiple correct answers (indicated).

Question 1: Inference Objective [Conceptual]

When performing inference in a Bayesian network to compute $P(\text{Query} \mid \text{Evidence})$, what is the primary computational task?

A. Finding the most likely value of the query variable

B. Marginalizing over hidden variables while conditioning on evidence

C. Building a new graphical model

D. Computing the prior probabilities of all nodes

E. Maximizing the joint probability distribution

Question 2: Identifying Hidden Variables [Conceptual]

Consider a Bayesian Network with variables $\{X, Y, Z, W, V\}$. For the query $P(X \mid W = \text{true}, V = \text{false})$, identify the hidden variables.

A. X, W, V

B. Y, Z

C. W, V

D. X, Y, Z

E. No hidden variables exist

Question 3: Enumeration Complexity [Computational]

A Bayesian network has 6 binary hidden variables. Using inference by enumeration, how many joint probability assignments must be evaluated?

A. 6

B. 12

C. 36

D. 64

E. 128

Question 4: Understanding the Enumeration Formula [Conceptual]

In the inference formula:

$$P(Q \mid e) = \frac{\sum_h P(Q, e, h)}{\sum_{q'} \sum_h P(q', e, h)}$$

What role does the denominator play?

A. It computes the prior probability of the query

B. It serves as a normalization constant to ensure probabilities sum to 1

C. It eliminates the hidden variables

D. It represents the conditional probability tables

E. It calculates the marginal probability of hidden variables

Question 5: Efficiency of Variable Elimination [Conceptual]

What makes Variable Elimination more efficient than direct enumeration for Bayesian network inference?

A. It uses approximation instead of exact computation

B. It avoids repeated computations by eliminating variables incrementally

C. It requires fewer CPT entries

D. It can skip hidden variables

E. It converts the DAG into a tree structure

Question 6: Understanding Factors [Conceptual]

In the context of Variable Elimination, what best describes a "factor"?

A. A directed edge in the Bayesian network

B. A table that assigns numerical values to variable assignments

C. The probability of observing evidence

D. A variable to be eliminated

E. The output of the inference algorithm

Question 7: Core Operations in VE [Conceptual]

What are the two essential operations used in Variable Elimination?

A. Union and Intersection

B. Factor Multiplication and Summation (Marginalization)

C. Addition and Normalization

D. Conditioning and Sampling

E. Join and Projection

Question 8: Result of Marginalization [Computational]

If we have a factor $g(X, Y, Z)$ and we marginalize out variable Y (compute $\sum_Y g(X, Y, Z)$), which variables remain in the resulting factor?

A. Only X

B. Only Y

C. Only Z

D. X and Z

E. X, Y, and Z

Question 9: Variable Elimination on a Chain [Critical Thinking]

Given the chain structure $X \to Y \to Z$, to compute $P(Z \mid X = \text{true})$, which variable must Variable Elimination eliminate?

A. X

B. Y

C. Z

D. All three variables

E. None, direct computation is possible

Question 10: Factor Product Computation [Computational]

Given $P(X)$ with $P(X = T) = 0.4$, and $P(Y \mid X)$ with $P(Y = T \mid X = T) = 0.9$ and $P(Y = T \mid X = F) = 0.3$. Calculate $P(X = T, Y = T)$.

A. 0.12

B. 0.27

C. 0.36

D. 0.40

E. 0.54

Question 11: Normalization Calculation [Computational]

Variable Elimination produces $P(B = T, e) = 0.12$ and $P(B = F, e) = 0.28$. What is $P(B = T \mid e)$?

A. 0.12

B. 0.28

C. 0.30

D. 0.40

E. 0.70

Question 12: Conditional Independence and Inference [Conceptual]

If variables X and Y are d-separated given Z in a Bayesian network, what inference simplification is possible?

A. $P(X \mid Y, Z) = P(X \mid Z)$ – knowing Y adds no information about X given Z

B. $P(X, Y) = P(X)P(Y)$ – unconditional independence

C. $P(Z \mid X, Y) = P(Z)$ – Z is independent of both

D. X and Y must share a common parent

E. The network can be simplified to a tree

Question 13: Importance of Elimination Order [Critical Thinking]

In Variable Elimination, why is the order in which variables are eliminated important?

A. Different orders produce different final answers

B. The order determines the size of intermediate factors, affecting computational cost

C. Some orders may cause the algorithm to fail

D. The order must match the topological sort of the DAG

E. It changes the conditional independence relationships

Question 14: Diagnostic Reasoning [Critical Thinking / Computation]

A cybersecurity system has:

- $P(\text{Attack}) = 0.02$ (2% of traffic is malicious)
- $P(\text{Alert} \mid \text{Attack}) = 0.95$ (95% detection rate)
- $P(\text{Alert} \mid \neg\text{Attack}) = 0.08$ (8% false alarm rate)

Given an alert, what is approximately $P(\text{Attack} \mid \text{Alert})$?

A. About 19%

B. About 45%

C. About 75%

D. About 95%

E. Cannot be determined

Question 15: Variable Elimination Properties [Multi-Select]

Which of the following statements about Variable Elimination are TRUE? **(Select all that apply)**

A. VE always produces exact inference results

B. VE runs in polynomial time on all Bayesian networks

C. VE processes and removes one hidden variable at a time

D. VE only works on tree-structured networks

E. The choice of elimination order can significantly impact performance

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— End of Quiz —